While the string would still be free to vibrate, Polchinski and his collaborators showed that its endpoints would be "stuck" or "trapped" within certain regions.

In some situations, the region might be one-dimensional, in which case the string's endpoints would be like two beads sliding on a wire, with the string itself being like a cord connecting them. In other situations, the region might be two-dimensional, in which case the endpoints of the string would be very much like two pinballs connected by a cord, rolling around a pinball table. In yet other situations, the region might have three, four, or any other number of spatial dimensions less than ten. These results, as shown by Polchinski and also by Petr Hořava and Michael Green, helped resolve a long-standing puzzle in the comparison of open and closed strings, but over the years, the work attracted limited attention.[5] In October 1995, when Polchinski finished rethinking these earlier insights in light of Witten's new discoveries, that changed.

A question that Polchinski's earlier paper left without a complete answer is one that may have occurred to you while reading the last paragraph: If the endpoints of open strings are stuck within a particular region of space, *what is it that they are stuck to*? Wires and pinball machines have a tangible existence independent of the beads or balls that are constrained to move along them. What about the regions of space to which the endpoints of open strings are constrained? Are they filled with some independent and fundamental ingredient of string theory, one that jealously clutches open string endpoints? Prior to 1995, when string theory was thought to be a theory of strings only, there didn't seem to be any candidate for the job. But after Witten's breakthrough and the torrent of results it inspired, the answer became obvious to Polchinski: if the endpoints of open strings are restricted to move within some *p*-dimensional region of space, then that region of space must be occupied by a *p*-brane.[37] His calculations showed that the newly discovered *p*-branes had exactly the right properties to be the objects that exert an unbreakable grip on open string endpoints, constraining them to move within the *p*-dimensional region of space they fill.

To get a better sense for what this means, look at Figure 13.2. In (a), we see a couple of two-branes with a slew of open strings moving